

The empirical recurrence for OEIS A260498, recovered from its tail

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Abstract

OEIS A260498 records “Empirical recurrence of order 88” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 143$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 4$, so the recurrence is proved for every $n \geq 92$. Its last coefficient is $c_{88} = -3 \neq 0$, so no further tail satisfies anything shorter; any order-88 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A260498 is “Number of $(n+2) \times (5+2)$ 0..1 arrays with each 3×3 subblock having clockwise perimeter pattern 00000001 00000011 or 00001011.”. It has offset 1 and begins

722, 942, 2420, 4471, 5848, 13651, 30590, 44345, ...

The entry states:

Empirical recurrence of order 88 (see link above)

As of the “Last modified” line on the live entry (Jun 02 2025 12:08:12, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 143$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 143$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 88: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 4$ is the first whose minimal order is exactly the stated 88.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 286$ exact terms from $n = 4$ on, it returns order 88 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{88} c_i a(n-i),$$

c_1	0	c_{23}	−214450	c_{45}	−3332270	c_{67}	142748
c_2	0	c_{24}	−4855	c_{46}	−4440301	c_{68}	142732
c_3	12	c_{25}	339456	c_{47}	−3917680	c_{69}	−12570
c_4	24	c_{26}	586449	c_{48}	−947131	c_{70}	−18351
c_5	0	c_{27}	539398	c_{49}	2765098	c_{71}	−27290
c_6	−58	c_{28}	−43277	c_{50}	3663048	c_{72}	−39791
c_7	−252	c_{29}	−856726	c_{51}	3378684	c_{73}	900
c_8	−273	c_{30}	−1315622	c_{52}	1193254	c_{74}	1375
c_9	156	c_{31}	−1110422	c_{53}	−1799086	c_{75}	2636
c_{10}	1082	c_{32}	123146	c_{54}	−2376606	c_{76}	7936
c_{11}	2508	c_{33}	1666548	c_{55}	−2313184	c_{77}	24
c_{12}	1711	c_{34}	2371531	c_{56}	−1088697	c_{78}	60
c_{13}	−2604	c_{35}	1934172	c_{57}	888682	c_{79}	98
c_{14}	−9522	c_{36}	−134966	c_{58}	1179623	c_{80}	−1072
c_{15}	−15522	c_{37}	−2577206	c_{59}	1229336	c_{81}	−6
c_{16}	−6247	c_{38}	−3520295	c_{60}	739826	c_{82}	−14
c_{17}	20380	c_{39}	−2893372	c_{61}	−321378	c_{83}	−54
c_{18}	52545	c_{40}	−61122	c_{62}	−433497	c_{84}	87
c_{19}	66842	c_{41}	3239734	c_{63}	−493058	c_{85}	0
c_{20}	12439	c_{42}	4349790	c_{64}	−376391	c_{86}	0
c_{21}	−99594	c_{43}	3688772	c_{65}	80548	c_{87}	4
c_{22}	−203285	c_{44}	482703	c_{66}	112057	c_{88}	−3

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 92$, and it is the recurrence OEIS A260498 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 4$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{88} - c_1x^{88-1} - \dots - c_{88}$. Its constant term is $-c_{88} = 3$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 88 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 88, so $\deg q = 88$, and it is monic. It holds from some index t on. From $\max(t, 4)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 88, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 26 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 88, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -3 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-88 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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